MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**December, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Civil Engineering**	**Course Type**	**FC**
Course Name	**Precast Technology**	**Course Code**	**CIE 3125**
Semester	**V**	**Max Marks**	**50**
Date		**Duration**	**3 hrs**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	In a precast concrete project, a series of wall panels are being manufactured off-site and will be transported to the construction site for installation. The design specifies that the panels should have a maximum variation tolerance of ±1/4 inch in height and width. During the quality control process, it is discovered that several panels have a height variation of up to ±3/8 inch and width variation of ±1/2 inch. Discuss the potential impacts of these variations on the overall project, including structural integrity, aesthetics, and installation challenges.	**4**	3	1,2,4,5,7,10,11, PSO2	**3**
1B	Identify the challenges involved in adopting the precast foundation over cast in situ foundation.	**3**	3	1,2,4,5,7,10,11, PSO2	**2**
1C	Summarize the advantages of pre-stressed concrete over ordinary reinforced cement concrete.	**3**	2	1,4,7,12,PSO2	**2**
2A	A new primary school was planned in a suburban area to accommodate increasing student enrollment. The project aimed to create a modern educational environment while minimizing	**4**	3	1,2,4,5,7,10,11, PSO2	**3**

	construction time and disruption. Which type of prefabrication system would you choose for the given case? Appraise the suitability of the prefabrication system considering the scope of the project.				
2B	Identify the factors influencing the design of the precast concrete elements.	**4**	3	1,2,4,5,7,10,11, PSO2	**2**
2C	How can ferrocement be effectively applied in the construction of public toilets in India to address challenges such as affordability, durability, and rapid construction?	**2**	1	1,11,12,PSO2	**3**
3A	Sketch the standard precast concrete beam and girder shapes.	**4**	3	1,2,4,5,7,10,11, PSO2	**2**
3B	Consider the following case: The construction of a bridge to improve connectivity between rural and semi urban area in southern part of India. The client and the contractor has decided to adopt site prefabrication type over factory prefabrication. Justify the selection of the manufacture method.	**3**	4	1,4,5,6,9,11,12, PSO2	**3**
3C	Distinguish between stand method and flow method of manufacture of precast units.	**3**	4	1,4,5,6,9,11,12, PSO2	**2**
4A	Breakdown the processes involved in the stages of precasting concrete products.	**4**	4	1,4,5,6,9,11,12, PSO2	**2**
4B	Identify the suitable precasting method for the following elements. List the advantages of using the specific method for production. a) Double tees b) Hollow core slab	**3**	4	1,4,5,6,9,11,12, PSO2	**3**
4C	Describe the following methods of accelerated hardening of precast elements: a) Heating the aggregates and water b) Electrical method	**3**	4	1,4,5,6,9,11,12, PSO2	**2**
5A	Sketch the column to foundation connection showing the following stages: Before assembly, after assembly showing various details.	**4**	5	1,2,11,12,PSO2	**2**
5B	Identify the types of connections available for joining several precast elements.	**3**	5	1,2,11,12,PSO2	**2**

5C	Identify the possible causes for cracking of precast elements.	**3**	4	1,4,5,6,9,11,12, PSO2	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	2
2	3
3	19
4	19
5	7

BL	Total Marks
2	**34**
3	**16**
4	
5	